

Theorem 3. The period transducer

The defect chain $\delta \rightarrow \gamma \rightarrow \delta_{\text{eff}} \rightarrow \Delta_{\text{Ch}}$ and the modulation map (ϵ, ν, C_N)

Isaev Iskhak Khamzatovich (ORCID 0009-0003-7299-0701)

The cycloring research program · v1.0.0 · Theorem Monograph Edition

Study	CYCLORING
Study	CYCLORING — the Gamma-periods, the defect chain, and the synchronous breathing of a vortex ring
Author	Isaev Iskhak Khamzatovich (ORCID 0009-0003-7299-0701)
Version	1.0.0 — the theorem monograph edition
Date	2026-10-08
Code	cycloring/python/cycloring/ (periods · chain · ring · dynamics · ladder · runner · figures)
Data	cycloring/results/protocols/W1..W7_default.json
License	LicenseRef-Proprietary-Wild8Highlander-1.0

Table of Contents

Abstract.....	
Abstract.....	1
At a glance.....	1
1. The defect chain.....	1
2. The modulation map.....	2
3. Monotonicity and the limit.....	2
4. The stiffness power law.....	2

Note: this Table of Contents is generated via field codes. After editing, right-click the TOC and select "Update Field" to refresh the page numbers.

Abstract

The monograph builds a finite chain converting the arithmetic of the cyclotomic level into the modulation program of the ring. The input is the triple (N, B, λ_0) : the lattice order, the angular impulse of the frozen ring $B = N \Gamma R_0^2$, and its frozen frequency $\lambda_0 = \Gamma(N-1)/(4 \pi R_0^2)$. The chain: $\delta = \pi/N$, $k = \text{ceil}(B \lambda_0 / \Gamma^2)$, $\gamma = \delta^4/k$, $\delta_{\text{eff}} = \delta^5/k$, $\Delta_{\text{Ch}} = \gamma W_N/(N-1)$.

The transducer converts the discriminant Δ_{Ch} into the breathing program: $\epsilon = \Delta/(1+\Delta)$, $\nu = \omega_L (1+\Delta)^3/(1+2 \Delta)^{3/2} = \omega_L (1-\epsilon^2)^{-3/2}$, $C_N = \ln(1+1/\Delta) = -\ln \epsilon$. Well-definedness, bijectivity, the frequency-branch identity, the strict monotonicity of the defect $\gamma(N)$, and the classical limit are proved: $\Delta \rightarrow 0$ recovers the rigid ring.

The registered table of the four levels (7, 9, 15, 30): the breathing amplitude falls from $3.664\text{e-}3$ to $2.135\text{e-}7$ — the high levels are stiff. On the grid $N = 3..30$ the discriminant follows the power law $\Delta_{\text{Ch}}(N)$ of about $1.7\text{e}3 N^{-6.7}$ (a numerical approximation).

At a glance

Input	the triple $(N, B, \lambda_0) = (\text{level}, N \Gamma R_0^2, \Gamma(N-1)/(4 \pi R_0^2))$
Stiffness index	$k = \text{ceil}(B \lambda_0 / \Gamma^2) = \text{ceil}(N(N-1)/(4 \pi))$
Chain	$\delta = \pi/N$; $\gamma = \delta^4/k$; $\delta_{\text{eff}} = \delta^5/k$; $\Delta_{\text{Ch}} = \gamma W_N/(N-1)$
Amplitude	$\epsilon = \Delta/(1+\Delta)$ in $(0,1)$; the inverse $\Delta = \epsilon/(1-\epsilon)$
Frequency	$\nu = \omega_L (1+\Delta)^3/(1+2 \Delta)^{3/2} = \omega_L (1-\epsilon^2)^{-3/2}$
Stiffness	$C_N = \ln(1+1/\Delta) = -\ln \epsilon$

1. The defect chain

The level triple consists of the exact integral registers of the ring: $B = N \Gamma R_0^2$ — the angular impulse, $\lambda_0 = \Gamma(N-1)/(4 \pi R_0^2)$ — the frozen frequency. Their product is dimensionless and defines the stiffness index $k = \text{lceil } B \lambda_0 / \Gamma^2 \text{rceil}$; the defects then cascade: $\gamma = \delta^4/k$, $\delta_{\text{eff}} = \delta^5/k = \delta \gamma$, and the periodic completion is $\Delta_{\text{Ch}} = \gamma W_N/(N-1)$.

$$\delta = \frac{\pi}{N}, \quad k = \left\lceil \frac{B\lambda_0}{\Gamma^2} \right\rceil, \quad \gamma = \frac{\delta^4}{k}, \quad \delta_{eff} = \frac{\delta^5}{k}, \quad \Delta_{Ch} = \frac{\gamma W_N}{N-1}$$

2. The modulation map

The transducer is three formulas: $\varepsilon = \Delta/(1+\Delta)$; $\nu = \omega_L((1+\Delta)^3/((1+2\Delta)^{3/2}))$; $C_N = \ln(1+1/\Delta)$. The frequency-branch identity is the core of the theorem: from $\varepsilon = \Delta/(1+\Delta)$ follows $1 - \varepsilon^2 = (1+2\Delta)/((1+\Delta)^2)$, hence $\nu = \omega_L(1-\varepsilon^2)^{-3/2}$ — the transducer output coincides with the synchronous law of Theorem 4.

$$\varepsilon = \frac{\Delta}{1+\Delta}, \quad \nu = \omega_L \frac{(1+\Delta)^3}{(1+2\Delta)^{3/2}} = \omega_L(1-\varepsilon^2)^{-3/2}, \quad C_N = -\ln \varepsilon$$

3. Monotonicity and the limit

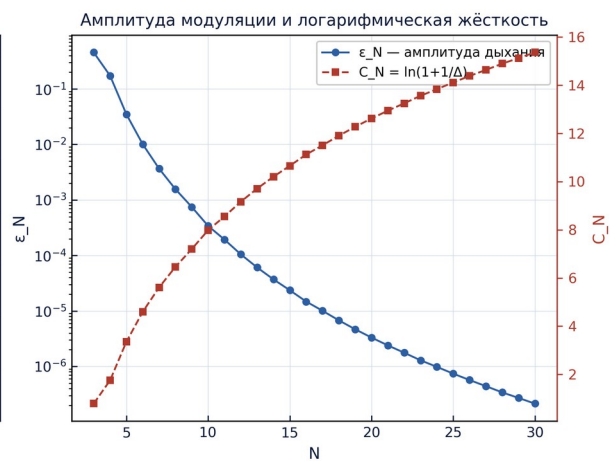
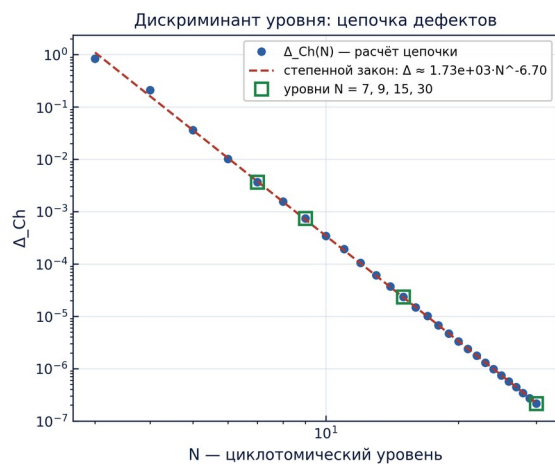
The defect $\gamma = \delta^4/k$ strictly decreases in N : the numerator π^4/N^4 falls, the denominator $k = \lceil N(N-1)/(4\pi) \rceil$ strictly grows for $N \geq 4$ (the step $2N/(4\pi) > 1$). The classical limit: $\Delta \rightarrow 0$ gives $\varepsilon \rightarrow 0$, $\nu \rightarrow \omega_L$ (the expansion $(1-\varepsilon^2)^{-3/2} = 1 + (3/2)\varepsilon^2 + O(\varepsilon^4)$) and $C_N \rightarrow \infty$ — in the limit the transducer switches the breathing off, restoring the rigid ring.

The registered table of the levels (protocol W7, default)

N	k	W_N	Delta_Ch	eps_N	nu/ omega_L	C_N
7	4	2.175444	3.677e-03	3.664e-03	1.0000201	5.609
9	6	2.416543	7.474e-04	7.469e-04	1.0000042	7.200
15	17	2.915447	2.357e-05	2.357e-05	1.0000000	10.656
30	70	3.603649	2.135e-07	2.135e-07	1.0000000	15.360

4. The stiffness power law

On the grid $N = 3..30$ the discriminant is approximated by the power law $\Delta_{Ch}(N) \approx 1.7 \cdot 10^3 \cdot N^{(-6.7)}$. The honest caveat: this is a numerical approximation, not an asymptotic theorem — only the monotonicity of the defect is proved. The economical mechanism of the observed stiffness: $W_N \approx \ln N + O(1)$ grows slowly, $\gamma \sim \delta^4/k$ falls fast, and their product decays like a power.



The level discriminant, the amplitude, and the stiffness (protocol W7).